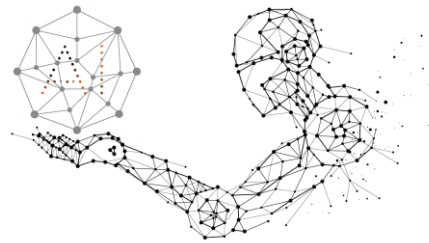
Applied Machine Learning

Chapter 3- Part 2- A Little Linear Algebra

دانشگاه تربیت مدرس
Tarbiat Modares University

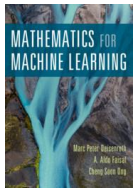
Hossein Homaei
Department of Electrical & Computer Engineering



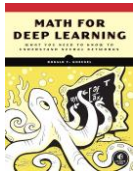


Some resources

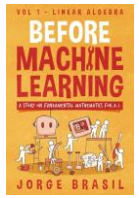
- Books



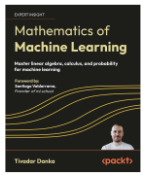
M. P. Deisenroth, A. A. Faisal, and C. S. Ong, *Mathematics for machine learning*. Cambridge University Press, 2020.



R. T. Kneusel, *Math for Deep Learning: What You Need to Know to Understand Neural Networks*. No Starch Press, 2022.



J. Brasil, *Before Machine Learning*, vol. 1, Linear Algebra. 2023.



T. Danka, *Mathematics of Machine Learning: Master Linear Algebra, Calculus, and Probability for Machine Learning*. Packt, 2025.

… Some resources

- Online
  - Linear Algebra for Machine Learning and Data Science
    - Instructor: Luis Serrano
    - DeepLearning.AI
    - **A significant portion of this lecture's content is sourced from this course.**
  - CS229: Machine Learning- The Summer Edition 2019
    - Instructor: Anand Avati
    - Stanford University
  - Essence of linear algebra
    - Grant Sanderson
    - 3Blue1Brown
  - Linear Algebra
    - Instructor: Gilbert Strang
    - MIT
  - Mathematic for AI [In Persian]
    - Instructor: Behrooz Nasihatkon
    - Khajeh Nasir Toosi University of Technology







Linear Algebra

- What is it about?
  - Manipulating vectors and matrices to do calculations
- Why important?
  - It is the backbone of many machine learning techniques
- What is a linear equation?
  - Any equation in this form: $a_1x_1 + a_2x_2 + \cdots + a_nx_n + b = 0$
  - Some examples of non-linear equations:
    - $x^2 + y^2 = 0$
    - $\sin(x) + y^2 = 12$
    - $2^x - 3^y = 0$



Solving system of linear equations



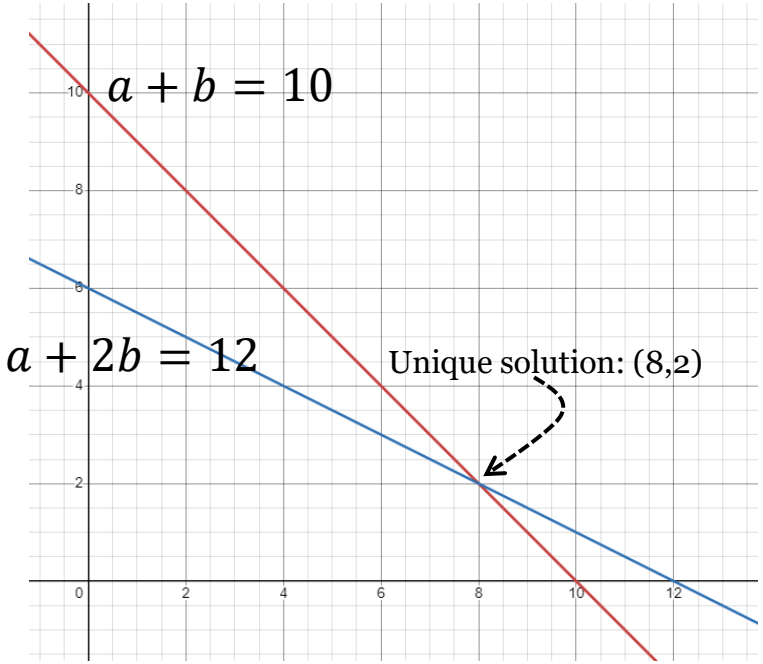
… Singularity concept

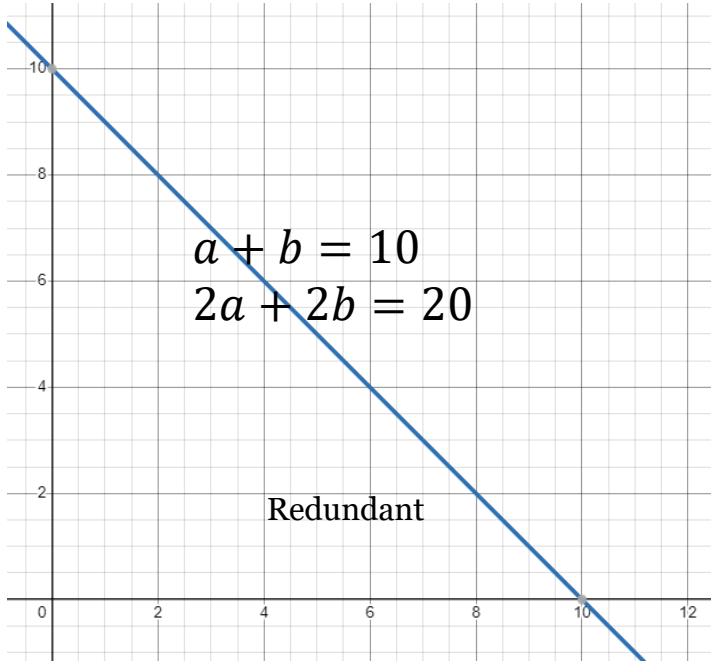
System 1	System 2	System 3
$a + b = 10$ $a + 2b = 12$	$a + b = 10$ $2a + 2b = 20$	$a + b = 10$ $2a + 2b = 24$
Complete system	Redundant system	Contradictory system
Unique solution	Infinitely many solutions	No solution
Non-singular	**Singular**	**Singular**

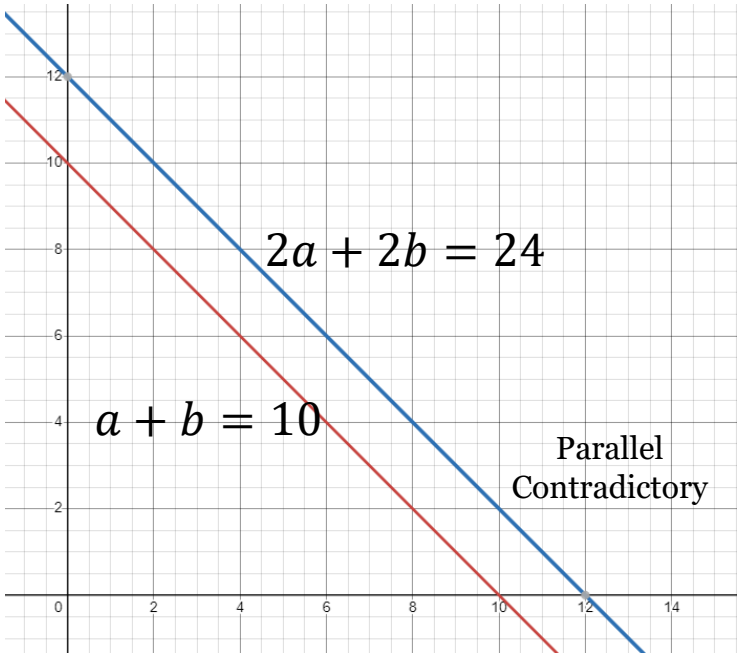


… Singularity concept

System 1	System 2	System 3
Non-singular	Singular	Singular







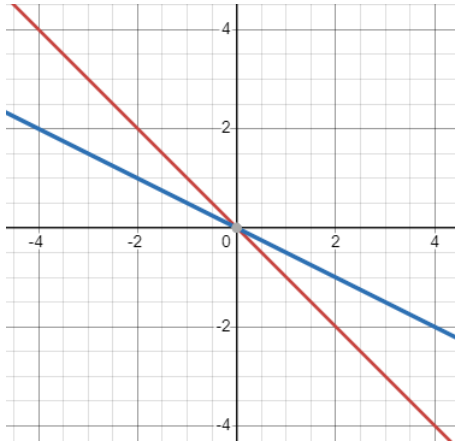
… Singularity concept

System 1

$a + b = 10 \rightarrow a + b = 0$

$a + 2b = 12 \rightarrow a + 2b = 0$

non-singular → non-singular

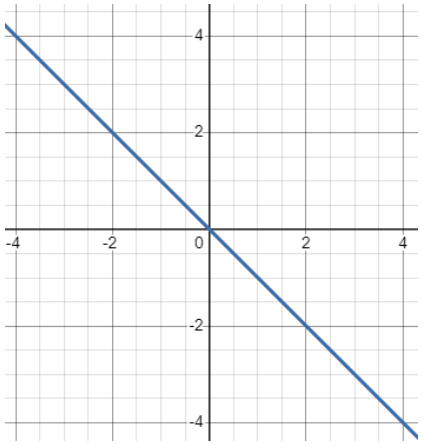


System 2

$a + b = 10 \rightarrow a + b = 0$

$2a + 2b = 20 \rightarrow a + b = 0$

singular → singular

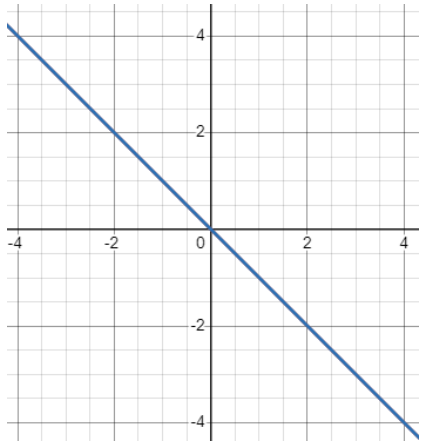


System 3

$a + b = 10 \rightarrow a + b = 0$

$2a + 2b = 24 \rightarrow a + b = 0$

singular → singular



Conclusion

Constants in the system don't matter in determining the singularity.

Matrix Singularity

System of equations	Corresponding matrix
$a + b = 0$ $a + 2b = 0$	a b 1 1 1 2
Non-singular system	Non-singular matrix

No row is a multiple of the other one

Rows are linearly independent
(columns are also linearly independent)

System of equations	Corresponding matrix
$a + b = 0$ $2a + 2b = 0$	a b 1 1 2 2
singular system	Singular matrix

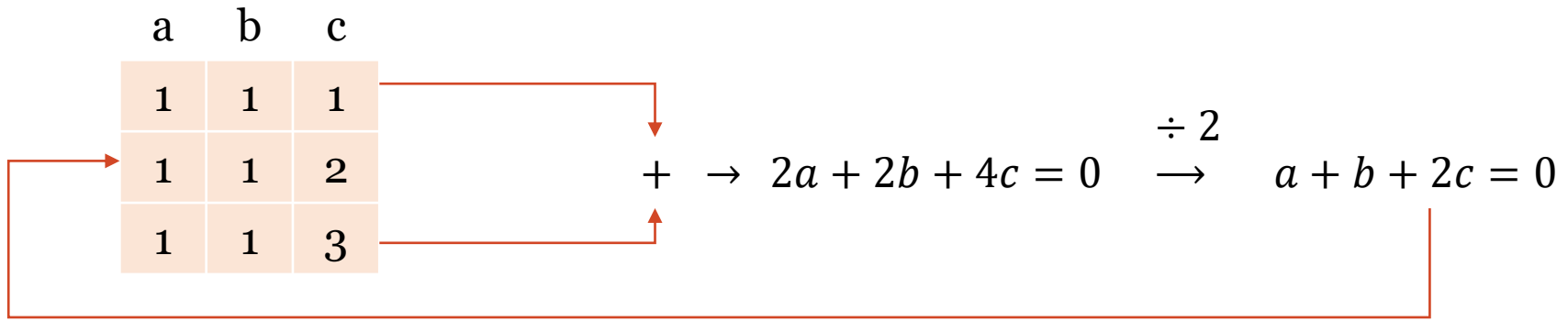
2nd row is a multiple of the 1st row

Rows are linearly dependent



… Matrix Singularity

$$a + b + c = 0$$
$$a + b + 2c = 0$$
$$a + b + 3c = 0$$



2nd row depends of the 1st and 3rd rows

Rows are linearly dependent
The matrix is singular

… Matrix Singularity

$$a + b + c = 0$$
$$a + 2b + c = 0$$
$$a + b + 2c = 0$$

a	b	c
1	1	1
1	2	1
1	1	2

There is no relation between rows and columns

Rows and columns are linearly independent
The matrix is non-singular
How can we determine that?



… Matrix Singularity

- Assume that a matrix is singular
  - The 2nd row depends on the 1st row

a	b
c	d

$$\begin{array}{|c|c|} \hline a & b \\ \hline \end{array} \times k = \begin{array}{|c|c|} \hline c & d \\ \hline \end{array} \rightarrow \begin{cases} ak = c \rightarrow k = \frac{c}{a} \\ bk = d \rightarrow k = \frac{d}{b} \end{cases} \rightarrow \frac{c}{a} = \frac{d}{b} \rightarrow ad = bc$$

$$\boxed{ad - bc} = 0$$

Determinant



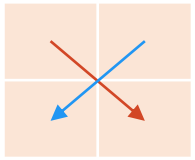
… Matrix Singularity

- We can decide about matrix singularity based on the determinant
  - Non-zero determinant → non-singular matrix
  - Zero determinant → singular matrix



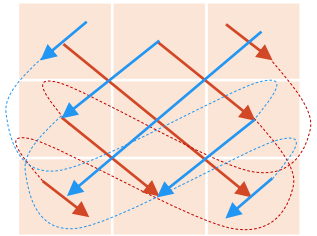
Determinant

- A method for computation



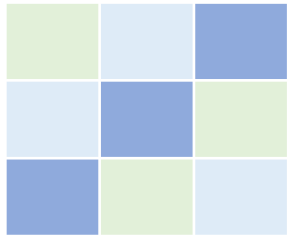


Add multiplication of  And subtract multiplication of 





Add multiplications of  , , and



And subtract multiplications of  , , and

… Determinant

- More formal formula (for 3×3 matrix)

$$\det\left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\right) = a \times \det\left(\begin{bmatrix} e & f \\ h & i \end{bmatrix}\right) - b \times \det\left(\begin{bmatrix} d & f \\ g & i \end{bmatrix}\right) + c \times \det\left(\begin{bmatrix} d & e \\ g & h \end{bmatrix}\right)$$



… Determinant

- More general: Laplace or cofactor expansion
  - The determinant for an $n \times n$ matrix A
    - The Laplace expansion along the ith row ($1 \leq i \leq n$) is

$$\det(A) = \sum_{j=1}^{n} (-1)^{i+j} a_{i,j} m_{i,j}$$

    - Where, $a_{i,j}$ is the entry of the ith row and jth column of A, and $m_{i,j}$ is the determinant of the submatrix obtained by removing the ith row and jth column of A

  - Usually, we set $i = 1$:

$$\det(A) = \sum_{j=1}^{n} (-1)^{j+1} a_{1,j} m_{1,j}$$



… Determinant

- Example-1

1	1	1
1	2	1
1	1	2

$$1 \times 2 \times 2 + 1 \times 1 \times 1 + 1 \times 1 \times 1 - 1 \times 2 \times 1 - 1 \times 1 \times 2 - 1 \times 1 \times 1$$
$$= 4 + 1 + 1 - 2 - 2 - 1 = 1$$

$$1 \times (4 - 1) - 1 \times (2 - 1) + 1 \times (1 - 2) = 3 - 1 - 1 = 1$$

The matrix is non-singular



… Determinant

- Example-2

2	1	3	4
0	-1	2	1
3	2	0	5
-1	3	2	1

$$2 \times \det(A_{11}) - 1 \times \det(A_{12}) + 3 \times \det(A_{13}) - 4 \times \det(A_{14})$$
$$= 80 + 10 + 57 - 112 = 35$$

The matrix is non-singular



Solving system of linear equations

Non-singular system

$$\begin{aligned} 5a + b &= 17 \\ 4a - 3b &= 6 \end{aligned}$$

Divide by coefficient of a →

$$\begin{aligned} a + 0.2b &= 3.4 \\ a - 0.75b &= 1.5 \end{aligned}$$

Subtract to eliminate a from one equation

$$0.95b = 1.9$$

$$b = 2$$

Solved system

$$a = 3$$



… Solving system of linear equations

Singular system

$$\begin{aligned} a + b &= 10 \\ 2a + 2b &= 20 \end{aligned}$$

Divide by coefficient of a →

$$\begin{aligned} a + b &= 10 \\ a + b &= 10 \\ \hline 0 &= 0 \end{aligned}$$

Subtract to eliminate a from one equation

$$a + b = 10$$

Solved system
a=x, b=10-x

The solution has **one degree of freedom**
If you vary x, you can get many different solutions to the system.

… Solving system of linear equations

Singular system

$$\begin{aligned} a + b &= 10 \\ 2a + 2b &= 24 \end{aligned}$$

Divide by coefficient of a →

$$\begin{aligned} a + b &= 10 \\ a + b &= 12 \end{aligned}$$

$$-$$

$$0 = 2$$

Subtract to eliminate a from one equation

↓

Contradiction

↓

There is no solution



… Solving system of linear equations

System with more variables

$$a + b + 2c = 12$$
$$3a - 3b - c = 3$$
$$2a - b + 6c = 24$$

The Divide by coefficient of $\boldsymbol{a}$

$$a + b + 2c = 12$$
$$a - b - \frac{1}{3}c = 1$$
$$a - \frac{1}{2}b + 3c = 12$$

Use the 1st equation to remove $\boldsymbol{a}$ from the others

$$a + b + 2c = 12$$
$$-2b - \frac{7}{3}c = -11$$
$$-\frac{3}{2}b + c = 0$$

Solve this new system of two equations

Divide by the coefficient of $\boldsymbol{b}$

$$b + \frac{7}{6}c = \frac{11}{2}$$
$$b - \frac{2}{3}c = 0$$

Use the 1st equation to remove $\boldsymbol{b}$ from the other

$$\frac{11}{6}c = \frac{11}{2} \rightarrow c = 3 \rightarrow b = 2 \rightarrow a = 4$$

Gaussian Elimination

- Solve a system of equations in matrix form
  - Same as we explained, but in a more simplified procedural form
  - Also called Row Reduction



… Gaussian Elimination

Example

Original system

$$5a + b = 17$$
$$4a - 3b = 6$$

Intermediate system

$$a + 0.2b = 3.4$$
$$b = 2$$

Solved system

$$a = 3$$
$$b = 2$$

Original matrix

5	1
4	-3

row manipulation

Upper diagonal matrix

1	0.2
0	1

Row Echelon Form

More manipulation

Diagonal matrix

1	0
0	1

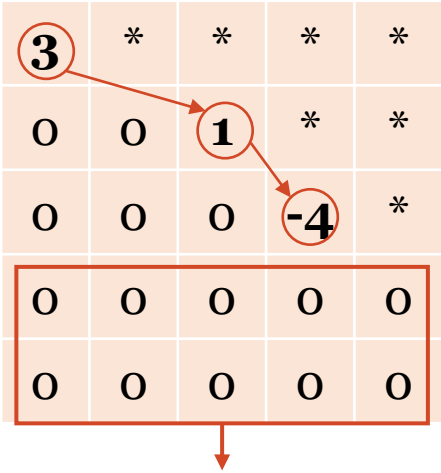
Reduced Row Echelon Form



… Gaussian Elimination

- Row Echelon Form
  - Potential zero rows at the bottom
  - The left-most non-zero entry (pivot) of every row is on the right of the leading entry of every row above.
    - So, Below the diagonal, everything is zero

Every pivot is to the right of the pivots on the rows above



Potential zero rows at the bottom

… Gaussian Elimination

- Reduced Row Echelon Form
  - Row Echelon Form
  - The pivot in each row is 1
    - Called leading one
  - Each column containing a leading 1 has zeros in all entries above the leading 1

Row echelon form

3	*	*	*	*
0	0	**1**	*	*
0	0	0	**-4**	*
0	0	0	0	0
0	0	0	0	0

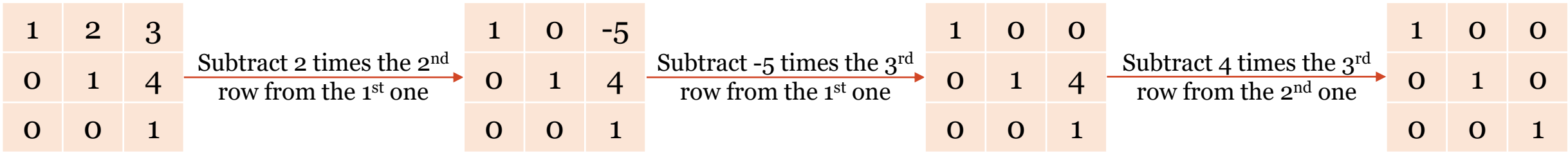
Reduced row echelon form

1	*	**0**	**0**	*
0	0	**1**	**0**	*
0	0	0	**1**	*
0	0	0	0	0
0	0	0	0	0



… Gaussian Elimination

- Transform Row Echelon to Reduced Row Echelon Form
  - Divide each row by its pivot
  - Clear out any number above the pivot



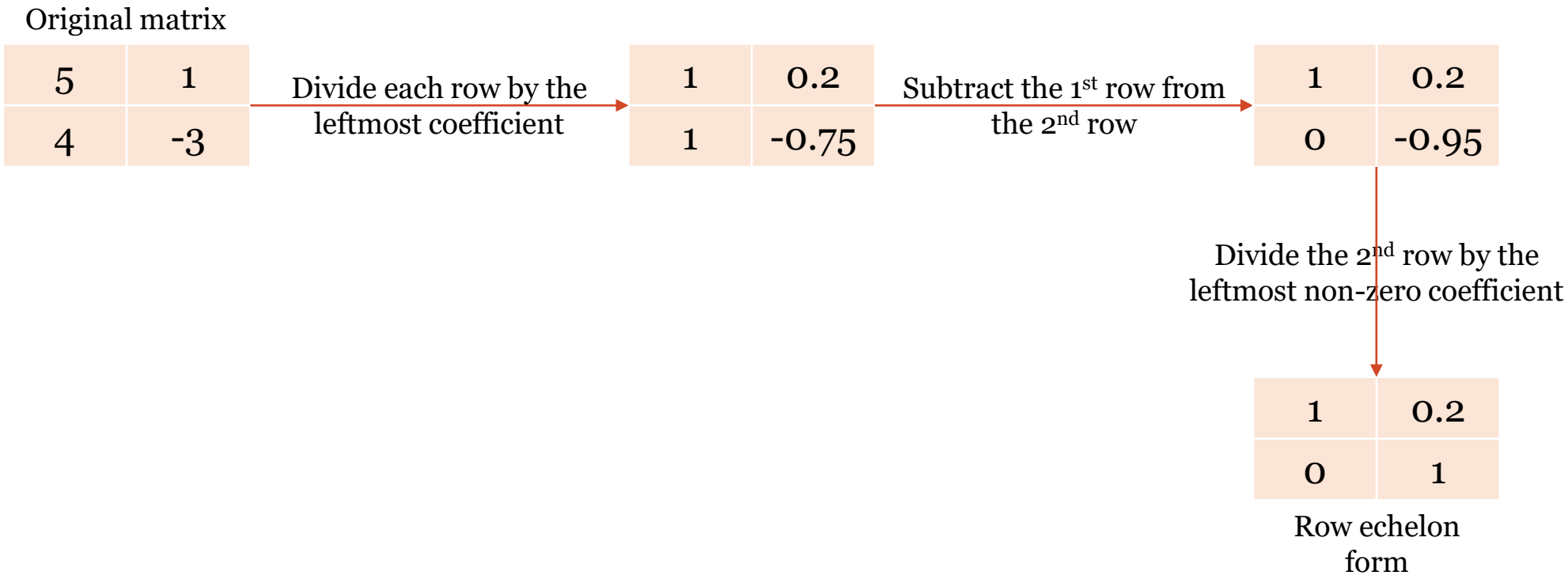
… Gaussian Elimination

- Row operations that preserve singularity
  - Switching rows
  - Multiplying a row by a non-zero scalar
  - Adding a row to another row



… Gaussian Elimination

- Example



… Gaussian Elimination

- Create the augmented matrix
  - Add constants of the equations to the right of the corresponding matrix
- Transform the matrix to the Reduced Row Echelon Form
  - Select the top left cell as the pivot
  - Use row operations to set your pivot to one
  - Use row operations to set all values below the pivot to zero
  - Repeat the above process row by row
- Solve the system using the back substitution process
  - Go from bottom to top rows
  - Use the pivot of each row to cancel the values in the cells above it



… Gaussian Elimination

- Example: non-singular

$$\begin{cases} 2a - b + c = 1 \\ 2a + 2b + 4c = -2 \\ 4a + b = 4 \end{cases}$$

R_1	2	-1	1	1
R_2	2	2	4	-2
R_3	4	1	0	4

$$\begin{bmatrix} R_1 \leftarrow \frac{1}{2}R_1 \\ R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - 4R_1 \\ R_2 \leftarrow \frac{1}{3}R_2 \\ R_3 \leftarrow R_3 - 3R_2 \\ R_3 \leftarrow \frac{-1}{5}R_3 \end{bmatrix}$$

R_1	1	-1/2	1/2	1/2
R_2	0	1	1	-1
R_3	0	0	1	-1

$$\begin{bmatrix} R_2 \leftarrow R_2 - R_3 \\ R_1 \leftarrow R_1 - \frac{1}{2}R_3 \\ R_1 \leftarrow R_1 + \frac{1}{2}R_2 \end{bmatrix}$$

	1	0	0	1
	0	1	0	0
	0	0	1	-1

Identity matrix

$$\begin{cases} a = 1 \\ b = 0 \\ c = -1 \end{cases}$$

Rank

System 1	System 2	System 3
The dog is black. The cat is orange.	The dog is black. The dog is black.	The dog! The dog!
Two sentences Two pieces of information	Two sentences One piece of information	Two sentences Zero piece of information
Rank = 2	**Rank = 1**	**Rank = 0**



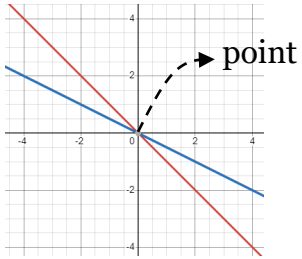
… Rank

Non-singular system

$$a + b = 0$$
$$a + 2b = 0$$

Two equations

Two pieces of information



Dimension of solution space= 0

1	1
1	2

Rank = 2

Full rank matrix

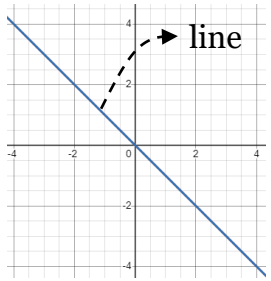
The rank is equal to the number of rows

singular system

$$a + b = 0$$
$$2a + 2b = 0$$

Two equations

One piece of information



Dimension of solution space= 1

1	1
2	2

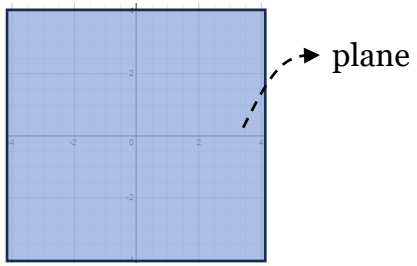
Rank = 1

singular system

$$0a + 0b = 0$$
$$0a + 0b = 0$$

Two equations

Zero pieces of information



Dimension of solution space= 2

0	0
0	0

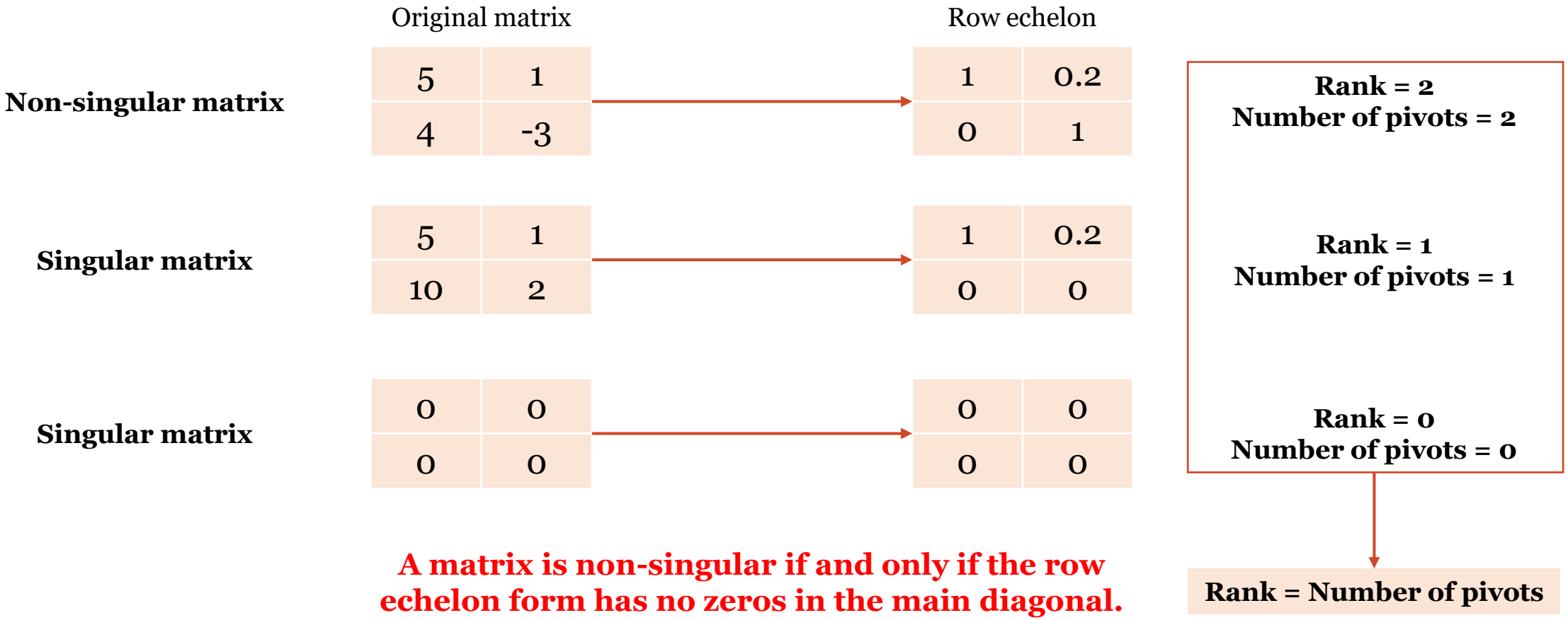
Rank = 0

Rank = (number of rows)-(dimension of solution space)

A matrix is non-singular if and only if it has full rank.

… Rank

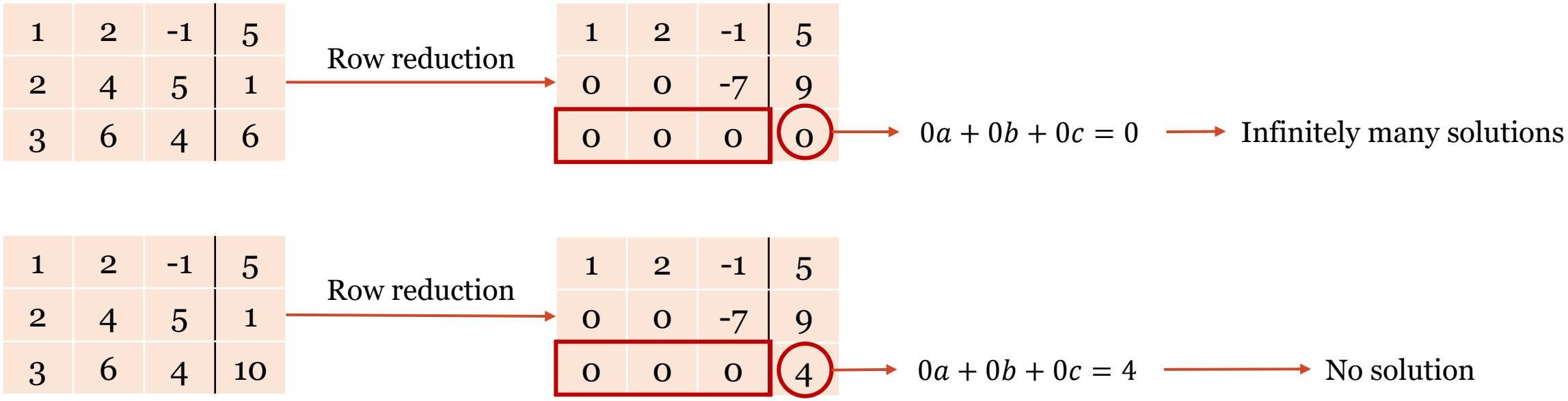
- Is there an easier way to calculate the rank?



A matrix is non-singular if and only if the row echelon form has no zeros in the main diagonal.

Gaussian Elimination (continue!)

- Example: Singular



Vectors and Matrices



Vector

- Vector is a column of numbers
  - Number of coordinates in the vector is the dimension of the space it leaves
    - Example: (4, 3) lives in the plane and (4, 3, 1) lives in the space

Notations

Row vector

$$(x_1 \quad x_2 \quad \dots \quad x_n)$$

$$[x_1 \quad x_2 \quad \dots \quad x_n]$$

Column vector

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Letters

$$\vec{x}$$

$$\boldsymbol{x}$$



Matrix

- Matrix is an array of numbers

Notation

$$\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$$



Vector Components

- Size
  - L1-norm

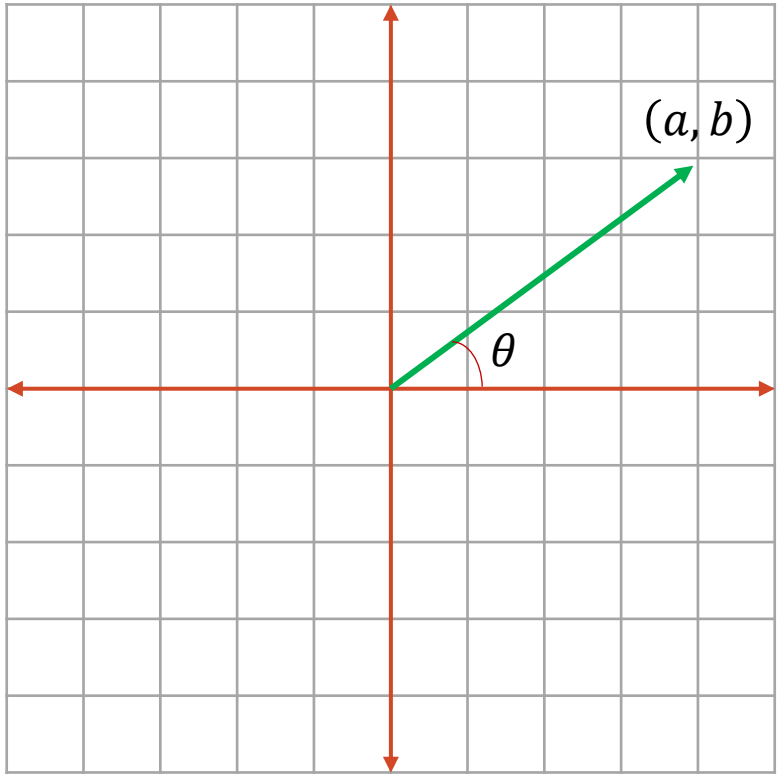
    $\|\vec{x}\|_1 = |x_1| + |x_2| + \cdots + |x_n|$

    Example: $|(a, b)|_1 = |a| + |b|$
  - L2-norm (default)

    $\|\vec{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$

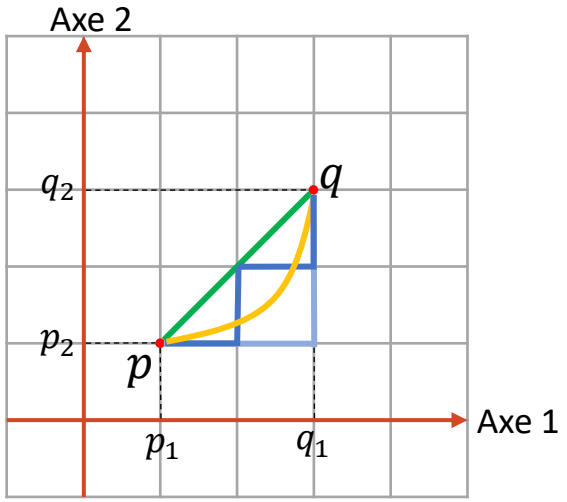
    Example: $|(a, b)|_2 = \sqrt{a^2 + b^2}$
- Direction

$$\tan\theta = \frac{b}{a} \rightarrow \theta = \arctan\left(\frac{b}{a}\right)$$



Distance

- The most common methods to calculate the distance between two points in an n-dimension space:
  - Euclidean distance: $\sqrt{\sum_{i=1}^{n}(q_i - p_i)^2}$
  - Manhattan distance: $\sum_{i=1}^{n}|q_i - p_i|$
  - Minkowski distance: $\sqrt[p]{\sum_{i=1}^{n}|q_i - p_i|^p}$
    - General form of distance
      - If $p = 1 \rightarrow$ Manhattan
      - If $p = 2 \rightarrow$ Euclidean
      - If $p = \infty \rightarrow$ Chebyshev
        - $\lim_{p\to\infty}(\sum_{i=1}^{n}|q_i - p_i|^p)^{\frac{1}{p}} = \max_{1\le i\le n}|q_i - p_i|$



Operations

- Assume
  - $\vec{u} = (u_1 \quad u_2 \quad \dots \quad u_n)$ is a vector
  - $\vec{v} = (v_1 \quad v_2 \quad \dots \quad v_n)$ is a vector
  - λ is an scalar

- Sum and subtract between vectors

$$\vec{u} + \vec{v} = (u_1 + v_1 \quad u_2 + v_2 \quad \dots \quad u_n + v_n)$$
$$\vec{u} - \vec{v} = (u_1 - v_1 \quad u_2 - v_2 \quad \dots \quad u_n - v_n)$$

- Scalar-vector multiplication

$$\lambda\vec{x} = (\lambda u_1 \quad \lambda u_2 \quad \dots \quad \lambda u_n)$$



… Operations

- Dot product between two vectors

$$\vec{u}.\vec{v} = u_1 \times v_1 + u_2 \times v_2 + \cdots + u_n \times v_n$$

  - Another notation: $\langle \vec{u}, \vec{v} \rangle$
  - It is common to represent the first and second vectors in the row and column formats, respectively.

$$[u_1 \quad u_2 \quad \dots \quad u_n].\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 \times v_1 + u_2 \times v_2 + \cdots + u_n \times v_n$$

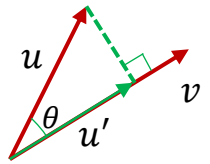


… Operations

- … Dot product
  - Connection between dot product and norm

$$\|\vec{u}\|_2 = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2} = \sqrt{u_1 \times u_1 + u_2 \times u_2 + \cdots + u_n \times u_n} = \sqrt{\langle \vec{u}, \vec{u} \rangle}$$

  - Connection between dot product and the angle between two vectors



$$\langle \vec{u}, \vec{v} \rangle = |u| \times |v| \times \cos(\theta) = |u'| \times |v|$$

    - Two vectors are orthogonal, if and only if their dot product is zero

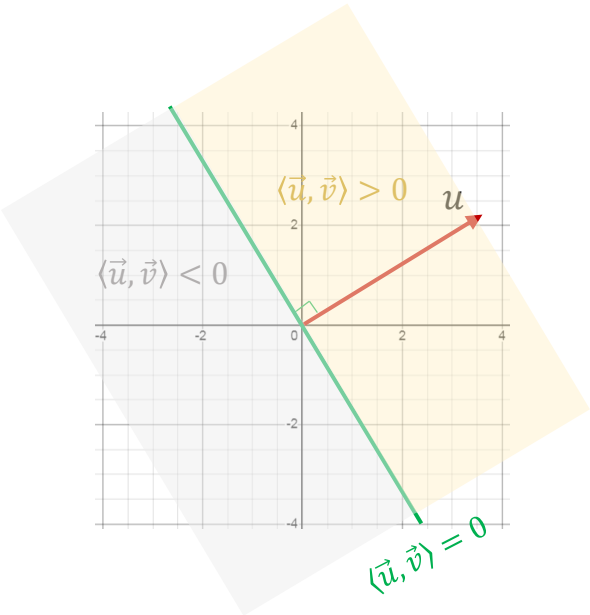
$$\langle \vec{u}, \vec{v} \rangle = 0 \rightarrow \vec{u} \text{ and } \vec{v} \text{ are orthogonal}$$

    - If the vectors are in the same direction

$$\langle \vec{u}, \vec{v} \rangle = |u| \times |v|$$

    - The dot product of a vector and itself

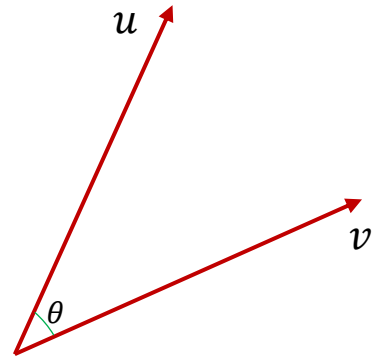
$$\langle \vec{u}, \vec{u} \rangle = |u|^2$$



… Operations

- … Dot product
  - Cosine Similarity
    - The cosine of the angle between the vectors

$$\cos\theta = \frac{\vec{u}.\vec{v}}{\|\vec{u}\|\|\vec{v}\|}$$



- Orthogonal vectors have no similarity
- Vectors with the same direction are very similar ($\cos\theta = 1$)

… Operations

- Vector transpose
  - Transforms column to the row or vice versa

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \rightarrow \vec{u}^T = [u_1 \quad u_2 \quad \dots \quad u_n]$$

$$\vec{u} = [u_1 \quad u_2 \quad \dots \quad u_n] \rightarrow \vec{u}^T = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

- Matrix transpose

$$\begin{pmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{pmatrix}^T_{m \times n} = \begin{pmatrix} x_{11} & \cdots & x_{m1} \\ \vdots & \ddots & \vdots \\ x_{1n} & \cdots & x_{mn} \end{pmatrix}_{n \times m}$$



… Operations

- Multiplication

$$\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}_{m\times n} \times \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1p} \\ y_{21} & y_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & x_{np} \end{bmatrix}_{n\times p} = \begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1p} \\ z_{21} & z_{22} & \cdots & z_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1} & z_{m2} & \cdots & z_{mp} \end{bmatrix}_{m\times p}$$

Where,

$$z_{ij} = x_{i1}y_{1j} + x_{i2}y_{2j} + \cdots + x_{in}y_{nj} = \sum_{k=1}^{n} x_{ik}y_{kj}$$

… Operations

- Inverse
  - Identity matrix
    - All elements on the diagonal = one & Any other element = zero

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

    - Why called identity?
      - Multiplying it by any vector results in the same vector
  - The inverse matrix is the matrix for which the product of matrices is the identity matrix.

$$AA^{-1} = I$$



… Operations

- … Inverse
  - How to find the inverse of a matrix?
    - By solving the system of linear equations!
    - Example: What is the inverse of matrix $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$?

$$\begin{array}{|c|c|} \hline 3 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \times \begin{array}{|c|c|} \hline a & b \\ \hline c & d \\ \hline \end{array} = \begin{array}{|c|c|} \hline 1 & 0 \\ \hline 0 & 1 \\ \hline \end{array}$$

$$\begin{cases} 3a + c = 1 \\ 3b + d = 0 \\ a + 2c = 0 \\ b + 2d = 1 \end{cases} \xrightarrow{\text{Use the Gaussian Elimination method to solve}} \begin{cases} a = \frac{2}{5} \\ b = -\frac{1}{5} \\ c = -\frac{1}{5} \\ d = \frac{3}{5} \end{cases}$$



… Operations

- … Inverse
  - Some cases have a straightforward formula
  - Example: 2×2 matrix
    - If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
    - Proof as an exercise!



… Operations

- … Inverse
  - Which matrices have an inverse?
    - Non-singular matrices (determinant ≠ zero) are invertible

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{cases} a + c = 1 \\ b + d = 0 \\ 2a + 2c = 0 \\ 2b + 2d = 1 \end{cases}$$

Contradictory!
$a + c = 1$ and $2a + 2c = 0$

No solution!
The matrix is not invertible



… Operations

- Determinant properties
  - Determinant of matrix product: $\det(A \times B) = \det(A) \times \det(B)$
    - Result: singular or non-singular matrix multiplied by a singular matrix → singular matrix
  - Determinant of inverses: $\det(A^{-1}) = \frac{1}{\det(A)}$
    - Proof:

$$\det(A \times A^{-1}) = \det(A) \times \det(A^{-1}) \rightarrow \det(I) = \det(A) \times \det(A^{-1}) \rightarrow \det(A^{-1}) = \frac{1}{\det(A)}$$



… Operations

- Trace
  - The trace of a square matrix is the sum of the elements on its main diagonal
    - $tr(A) = \sum_{i=1}^{n} a_{ii}$
  - Properties
    - $tr(A + B) = tr(A) + tr(B)$
    - $tr(cA) = ctr(A)$
    - $tr(A) = tr(A^T)$

